Student Course Management System – Project Report

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1. Cover Page

Project Title:
Student Course Management System

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Course Name:
Foundations of Software Development

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GitHub link: https://github.com/Ahmedma1her/Student-Course-Management-System

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2. Introduction

This project presents a console-based Student Course Management System developed using C++. The system is designed to efficiently manage student records, including their personal information and enrolled courses.

The project focuses on applying Object-Oriented Programming (OOP) concepts, utilizing the Standard Template Library (STL), and implementing file handling to ensure data persistence.

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4. Detailed Project Description

The Student Course Management System allows users to perform essential operations such as adding, removing, searching, and displaying student data. It also supports enrolling students in multiple courses and viewing those courses.

The system uses a menu-driven interface where users interact with the application through simple numeric inputs.

To ensure data persistence, all student data is saved in a file (`students.txt`) and reloaded when the program starts.

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5. Project Beneficiaries

- Students learning programming and OOP concepts
- Beginners in C++ development
- Educational institutions (basic prototype systems)
- Developers practicing STL and file handling

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6. Detailed Analysis

Functional Requirements:

- Add student (ID, Name, GPA)
- Remove student
- Search for student
- Display all students
- Enroll student in courses
- Show courses for a student
- Sort students by GPA
- Save and load data

Non-Functional Requirements:

- Easy-to-use interface
- Efficient execution
- Data persistence using file storage
- Input validation and error handling

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7. Task Breakdown Structure

Task	Description
System Design	Define structure and classes
Core Implementation	Develop main features
STL Usage	Implement vector, set, algorithms
File Handling	Save/load data
Testing	Validate functionality
Documentation	Prepare report and README

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8. Time Plan

Task	Duration
Planning	1 day
Development	3 days
Testing	1 day
Documentation	1 day

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9. System Architecture and Design

The system follows a simple modular design:

- **Main Module:** Controls program execution and menu
- **Student Class:** Stores and manages student data
- **File Handling Module:** Responsible for saving and loading data

Design Principles:

- Encapsulation: Data is protected within classes
- Inheritance: Student class extends Person class
- STL Usage: Efficient data handling using vector and set

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10. Testing Scenarios and Results

Test Case	Input	Expected Result	Status
Add Student	Valid input	Student added	✔
Add Student	Duplicate ID	Error message	✔
Remove Student	Existing ID	Removed successfully	✔
Search Student	Invalid ID	Not found	✔
GPA Validation	GPA > 4	Rejected	✔

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11. End-User Guide

1. Run the program
2. Choose an option from the menu
3. Enter required data
4. View results instantly
5. Exit the program to save data automatically

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12. Conclusion

The project successfully demonstrates the implementation of a student management system using C++. It applies key programming concepts such as Object-Oriented Programming, STL, and file handling.

The system is functional, efficient, and can be extended into more advanced applications in the future.

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13. References

- C++ Official Documentation
- https://www.cplusplus.com
- Course Materials (Foundations of Software Development)

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14. Appendices

- Source Code (project.cpp)
- UML Diagram